

# ISO-1 weak isogeny classes: trace census and a proved 2-adic obstruction

Research evidence report

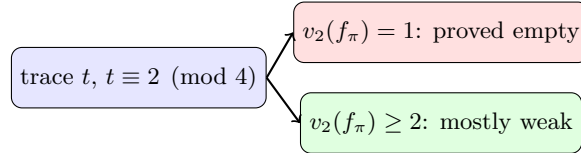
8 October 2026

**Requested scope.** Base primes  $p = 37$  through about 200,  $q = p^2$ , and curves over  $\mathbb{F}_{q^3} = \mathbb{F}_{p^6}$ : label every full-2-torsion isogeny class and find a criterion for weak-class reach. This round completed every-trace censuses at  $p = 37, 41, 43, 47$  and diagnostic censuses at  $p = 11, 13, 17$ . It proves a necessary class obstruction for every odd base prime. The other requested primes and an exact sufficient criterion remain open.

**Method.** A weak representative has  $y^2 = x(x - \alpha)(x - \sigma\alpha)$ , with  $\sigma$  the  $p^2$ -Frobenius. After translation and  $\mathbb{F}_{p^2}$  square scaling there are  $2q^2 + 2q$  normalized representatives. The  $p = 37$  run evaluated one representative per quadratic-twist pair and derived the opposite trace. Trace assignment uses a randomized two-point baby-step/giant-step counter. Its labels are computational evidence, not independent certificates of every cardinality. Full and twist-derived CSVs agreed exactly at  $p = 13$ ; two false individual assignments at  $p = 7$  were found by exact square-table counting. Independent PARI/GP `ellcard` checks placed 100, 3,952, 4,043, and 4,200 distinct norm-one Legendre traces at  $p = 37, 41, 43, 47$  in the respective positive sets (5,000 samples each at  $p = 41, 43, 47$ ).

| $p$       | $q$          | weak reps        | ordinary $t$   | weak $t$      | depth-1 weak    | depth- $\geq 2$ zero |
|-----------|--------------|------------------|----------------|---------------|-----------------|----------------------|
| 11        | 121          | 29,524           | 1,210          | 542           | 0/606           | 62/604               |
| 13        | 169          | 57,460           | 2,028          | 928           | 0/1,014         | 86/1,014             |
| 17        | 289          | 167,620          | 4,624          | 2,198         | 0/2,312         | 114/2,312            |
| <b>37</b> | <b>1,369</b> | <b>3,751,060</b> | <b>49,284</b>  | <b>24,352</b> | <b>0/24,642</b> | <b>290/24,642</b>    |
| <b>41</b> | <b>1,681</b> | <b>5,654,884</b> | <b>67,240</b>  | <b>33,224</b> | <b>0/33,620</b> | <b>396/33,620</b>    |
| <b>43</b> | <b>1,849</b> | <b>6,841,300</b> | <b>77,658</b>  | <b>38,452</b> | <b>0/38,830</b> | <b>376/38,828</b>    |
| <b>47</b> | <b>2,209</b> | <b>9,763,780</b> | <b>101,614</b> | <b>50,382</b> | <b>0/50,808</b> | <b>424/50,806</b>    |

At  $p = 37$  the run took 1,106.807 seconds with four Rayon threads and counted 563,856,133,786  $\mathbb{F}_p$  multiplications. The 50,654 CSV rows cover every candidate  $t \equiv 2 \pmod{4}$  in the Hasse interval; 49,284 are ordinary. Of 4,000 sampled random full-2-torsion curves, 2,433 were in an observed weak class: 0.6083, Wilson 95% interval [0.5930, 0.6233]. The zero-label share at  $p = 37$  is 50.59% of ordinary trace rows but 39.18% of sampled curves; these use different denominators. The  $p = 41, 43, 47$  sampled reach fractions are 0.6253, 0.6195, and 0.6320 (4,000 curves each; Wilson 95% intervals [0.6101, 0.6401], [0.6043, 0.6344], and [0.6169, 0.6468]). At  $p = 47$  the special size-two orbit yields the two nonordinary Hasse-boundary traces  $t = \pm 2 \cdot 47^3$  (two representatives each), independently confirmed with PARI/GP.



**Proof of the necessary obstruction.** Put  $K = \mathbb{F}_{p^6}$ ,  $Q = p^6$ , and  $\lambda = \alpha^{p^2}/\alpha$  on a weak model  $y^2 = x(x - \alpha)(x - \alpha^{p^2})$ . Then  $N_{K/\mathbb{F}_{p^2}}(\lambda) = 1$ , so  $\lambda$  has odd order dividing  $p^4 + p^2 + 1$ . Since  $p^2 \equiv 1 \pmod{4}$ , an explicit fourth root is  $\mu = \lambda^{(p^4+p^2+2)/4}$ , with  $\mu^4 = \lambda$ . The weak curve is a quadratic twist of  $L_\lambda : y^2 = x(x-1)(x-\lambda)$ . The rational 2-isogeny at  $(0, 0)$  has target

$$L'_\lambda : y^2 = x[x^2 + 2(1 + \lambda)x + (1 - \lambda)^2],$$

For  $x \neq 0$ , the map is

$$(x, y) \mapsto \left( x + a + \frac{b}{x}, y \left( 1 - \frac{b}{x^2} \right) \right), \quad a = -(1 + \lambda), \quad b = \lambda.$$

The identity and  $(0, 0)$  map to the identity. With  $U = x^2 + ax + b$ , clearing the curve equations gives  $U^2 - 2axU + (a^2 - 4b)x^2 = (x^2 - b)^2$ . The target roots are  $0, -(1 + \mu^2)^2, -(1 - \mu^2)^2$ . All their differences are squares: the final difference is  $\pm 4\mu^2$  and  $-1$  is a square in  $K$ . The roots are distinct since  $\lambda \neq 0, 1$ . By [Auer-Top, Lemma 2.1](#), each rational 2-torsion point has a rational half, and the halves of two independent points generate  $L'_\lambda[4]$ . Therefore  $16 \mid \#L_\lambda(K)$ , and the weak trace obeys  $t \equiv \pm(Q + 1) \pmod{16}$ .

Let  $t^2 - 4p^6 = f_\pi^2 D_K$ , where  $D_K$  is the fundamental CM discriminant. The measured  $v_2(f_\pi)$  is a class-wide bound, not a certified endomorphism-ring volcano level. For ordinary  $t \equiv 2 \pmod{4}$ , the arithmetic tests

$$v_2(f_\pi) \geq 2 \iff (t/2)^2 \equiv p^6 \pmod{16} \iff t/2 \equiv \pm p^3 \pmod{8}$$

are equivalent to the proved trace restriction. Indeed,  $d = (t/2)^2 - p^6$  has  $v_2(d) \geq 3$ . If  $v_2(d) = 3$ , then  $v_2(D_K) = 3$  and  $v_2(f_\pi) = 1$ ; if  $v_2(d) \geq 4$ , then  $v_2(t^2 - 4p^6) \geq 6$  and  $v_2(D_K) \in \{0, 2, 3\}$  forces  $v_2(f_\pi) \geq 2$ . Sufficiency fails on the recorded computational labels. There is a curve-level consequence: on the 2-isogenous neighbor with rational full 4-torsion,  $\pi - 1$  factors through the separable map [4]. Thus  $\eta = (\pi - 1)/4$  is an endomorphism, and  $\text{disc } \mathbb{Z}[\eta] = (t_L^2 - 4p^6)/16$ , where  $t_L$  is the untwisted Legendre trace. Its generated order has conductor  $f_\pi/4$ , so the neighbor's endomorphism conductor has 2-valuation at most  $v_2(f_\pi) - 2$ . Twisting preserves that order, and a degree-2 isogeny changes its 2-valuation by at most one (via the dual-isogeny inclusions of twice each endomorphism ring). Thus an ordinary weak curve satisfies  $v_2(f_{\text{End}(E)}) \leq v_2(f_\pi) - 1$ : it is never at the deepest possible 2-volcano level. Individual levels remain unmeasured. This bound also follows directly from full rational 2-torsion:  $(\pi - 1)/2$  is an endomorphism and its generated order has conductor  $f_\pi/2$ . It applies to every full-2-torsion curve in the population, so it does not further distinguish weak representatives. The extra class restriction comes from full 4-torsion on the neighbor. The accompanying identity records replay the target factorization and cleared isogeny equation at 32 recorded plus 64 additional points. Independent exact PARI/GP expansion checks both identities over the integers; the records cover those polynomial steps, while the theorem follows from the deduction above. The record IDs are `IDC1h8ae7209125dfe0a2` and `IDC1he02ab9a5cf019a56`.

**Held-out classifier.** On  $p = 37$  rows the necessary rule has 24,352 true positives, 290 false positives, no false negatives, and 24,642 true negatives (99.41% correct);  $p = 41, 43, 47$  add 396, 376, and 424 false positives, respectively. At  $p = 47$ , the necessary rule has 99.58% ordinary-row accuracy and no observed false negative.

**Counterexample to the proposed small invariants.** At  $p = 37$ ,  $t = -92,218$  has zero observed weak representatives and  $t = 38,854$  has 24. Both have Frobenius conductor depth 3, ramified 2, even maximal-order class number, and the same trace modulo  $2^{17}$ . Their PARI/GP class numbers are 912 and 3,168. Thus 2-splitting, conductor depth, small 2-power trace residues, and class-number parity do not classify all rows together. At  $p = 41$ , traces 136542 (zero) and 5470 (132 weak representatives) share depth 2, ramified 2, exact maximal-order class number  $h_K = 192$ , and the same trace modulo  $2^{17}$ . A near-edge pair, 133826 (zero) and 134082 (12), shares depth 3, split 2,  $h_K = 1680$ , and trace modulo 256. Class-number magnitude is associated with residual zeros, but does not yield an exact classifier even with these 2-adic features. At held-out  $p = 47$ ,  $t = 204238$  (zero) and  $t = 73166$  (96) likewise share depth 3, split 2, exact  $h_K = 336$ , and trace modulo  $2^{17}$ .

**Exact orbit reduction and audit.** Projective  $\alpha$  modulo  $\mathbb{F}_q^*$  maps bijectively to the norm-one parameters  $\lambda \neq 1$ . Their  $q$ -Frobenius conjugates and inverses have the same trace up to the quadratic twist already paired in the census. All such orbits have size six except the nontrivial cube roots of unity, which form one orbit of size two. Thus one point count per orbit suffices, reducing the present algorithm's point-count calls by nearly sixfold. At  $p = 13, 37$  the orbit CSVs match the prior direct CSVs byte for byte;  $p = 37$  needed 312,589 rather than 1,875,530 point counts. The corrected  $p = 37, 41, 43, 47$  counts obey the modulo-six constraint; at  $p = 47$  its special pair is nonordinary. This structural check is not an independent proof of every zero label.

**Where the residual zeros occur.** At  $p = 37$ , 234 of 290 high-depth zero rows lie in the outer fifth of  $|t|/(2p^3)$  (234/4,930 rows there, versus 6/4,926 in the inner fifth);  $p = 47$  has 392 of 424 residual zero rows in the outer fifth. This is an association, not a causal proof or an exact rule.

**Residual zeros and class-number magnitude.** The population is ordinary positive traces with  $v_2(f_\pi) \geq 2$ ; negative traces are their quadratic-twist mirrors. Native PARI/GP `qfbclassno` gives  $h_K$ . Entries are zero labels / all traces in each bin; the bins use  $q = p^2$ .

| $h_K/q$  | $p = 37$  | $p = 41$  | $p = 43$  | $p = 47$  | combined   |
|----------|-----------|-----------|-----------|-----------|------------|
| $< 1$    | 101/4,192 | 117/5,260 | 136/5,856 | 143/7,208 | 497/22,516 |
| [1, 2)   | 30/2,518  | 52/3,347  | 30/3,817  | 41/4,776  | 153/14,458 |
| [2, 4)   | 11/3,124  | 24/4,072  | 22/4,582  | 25/5,829  | 82/17,607  |
| [4, 8)   | 3/2,365   | 5/3,790   | 0/4,654   | 3/6,511   | 11/17,320  |
| $\geq 8$ | 0/122     | 0/341     | 0/505     | 0/1,079   | 0/2,047    |

These are complete observed strata, not sampled rates. The matched pairs above rule out using the listed features as an exact classifier of the recorded labels. Independent certification of every zero label remains open.

**Correction to prior evidence.** The earlier census used an  $\mathbb{F}_p$  nonsquare for the second  $\mathbb{F}_{p^2}$  square class. Every nonzero element of  $\mathbb{F}_p$  is a square in  $\mathbb{F}_{p^2}$ , so it duplicated one branch and omitted the other. The corrected code selects a genuine  $\mathbb{F}_{p^2}$  nonsquare. Old  $p \leq 31$  values remain historical;  $p = 23, 31$  await corrected reruns.

**Limit.** At  $p = 199$ , twist symmetry needs 1,568,278,802 point counts; the exact orbit quotient reduces this to 261,379,801. Scaling the measured  $p = 37$  direct run by work count and the approximate  $p^{3/2}$  cost per point count gives roughly 130 days for the older method on comparable resources. This is a crude extrapolation, not a run, and excludes growing trace-factorization work. The requested  $p = 53$ –200 census and a sufficient replacement formula have not been established. The theorem certifies absence in about half of the candidate classes and can safely prefilter them.

**Evidence.** The accompanying Markdown report links the code, complete CSVs, fit output, samples, and SVG. The  $p = 37$  command used `RAYON_NUM_THREADS=4` and `--derive-twists`. The weak-model and reach assumptions are in [Joux–Vitse, §4.1](#); [Auer–Top](#) proves a broader Legendre-family statement. The accompanying [Markdown report](#) links every CSV, the identity records, and the new validation receipt.

## Complete observed trace strata and invariant flow

### ISO-1: trace-level weak-class census

Ordinary full-2-torsion trace candidates over  $F_{(p^6)}$ ; corrected  $F_{(p^2)}$  square classes

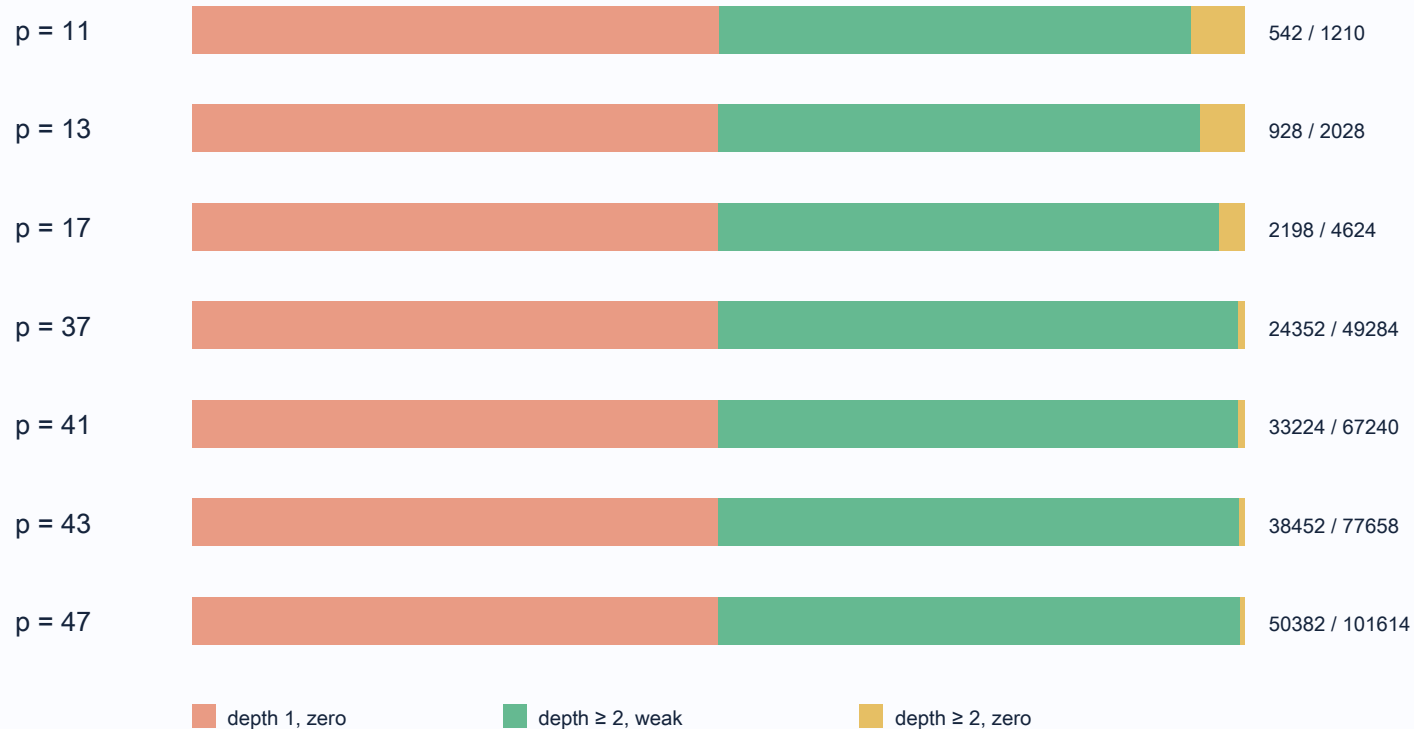
trace  $t$ ,  $t \equiv 2 \pmod{4}$   
class invariant:  $\Delta = t^2 - 4p^6$

$v_2(f_{\text{Frob}}) = 1$   
proved absent for every odd  $p$

$v_2(f_{\text{Frob}}) \geq 2$   
mostly weak; residual zero rows remain

Proof: norm-one  $\lambda \rightarrow$  fourth power  $\rightarrow$  2-isogenous full 4-torsion  $\rightarrow t \equiv \pm(p^6 + 1) \pmod{16}$

#### Class labels by stratum



Source: the seven census CSVs linked in the Markdown report, rendered by `iso1_class_census visual`. The bars count ordinary trace rows; their widths are fractions of each prime's row population. The right-hand labels give weak rows / ordinary rows. Sampling intervals apply to the separate 4,000-curve reach samples, not these census bars. The proof excludes every depth-1 class; the yellow segments retain the observed high-depth zero rows and their point-count uncertainty.